

Talos

Spoken instructions for a
physical dinner-table simulation

Two SO-101 arms
MuJoCo physics
Experimental learned bottle control



Intel online challenge + Speechmatics bonus

Dinner-table manipulation

“Set the table”

Place bottle, plate and mug.
Open the passive drawer.
Retrieve the fork and spoon.

Every grasp, lift and placement
passes physical contact checks.



Speech and control

Speechmatics streaming transcription

A short-lived browser token keeps the API key on the server.
Final transcripts become typed movement instructions.

A bounded language grammar selects a physical skill or relay.
The user chooses programmed or learned bottle control.

Visible task stages, cancellation and explicit failure feedback.

Two arms share the bottle

“Pass the bottle to the right arm”

- 1 Left arm lifts and releases in a shared table area.
- 2 Left arm parks before the right arm approaches.
- 3 Right arm regrips, lifts and places at the right side.

76.3 s simulated time 1.45 mm final placement error

Learned bottle controller

Overhead bottle localization and motor feedback

A classical RGB detector locates the bottle at startup.
A trained neural primitive predicts the joint trajectory.

OpenVINO generates joint targets.
MuJoCo actuators and finger contacts move the bottle.

An independent physical monitor checks support and placement.
The policy has no continuous visual correction yet.

Physical evaluation

Reliable task-preset starts, limited wider coverage

10 / 10

New task-preset scene seeds

1 / 3

Wider development starts

The ten-seed test uses small upright bottle-position changes.
Larger workspace moves and other learned skills remain unresolved.

Intel OpenVINO execution

0.157 ms

Intel CPU median inference

0.688 ms

Intel UHD iGPU median inference

FP32 model export with numerical parity checks.
Physical OpenVINO rollout passes familiar bottle starts.

Network timing only. NVIDIA renders the current camera images.
Core Ultra hardware execution has not been tested.

Reproducible prototype

Runnable simulation, weights and evidence

Six selectable views and four-view 720p recordings.
200 Hz physics control independent of browser frame rate.
Reproducible training and evaluation scripts.

Current scope: physical dinner skills, spoken commands,
a table-supported relay and a learned bottle baseline.

github.com/jannissio/talos-ai-infra